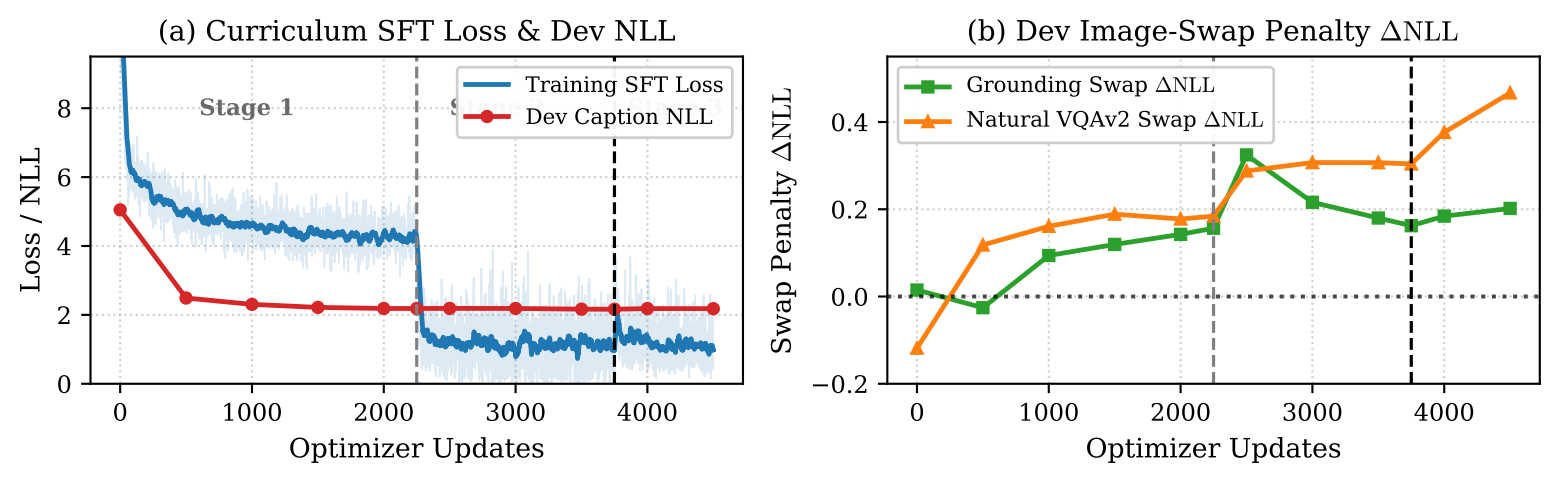

(a) Curriculum SFT Loss & Dev NLL
Stage 1
Training SFT Loss
Dev Caption NLL
Loss / NLL
Optimizer Updates
0
2
4
6
8
0
1000
2000
3000
4000
(b) Dev Image-Swap Penalty ΔNLL
Grounding Swap ΔNLL
Natural VQAv2 Swap ΔNLL
Swap Penalty ΔNLL
Optimizer Updates
−0.2
0.0
0.2
0.4
0
1000
2000
3000
4000